GRANT FLORES-DEWILDE

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EDUCATION

University of Pennsylvania | Ben Franklin Scholar | *Philadelphia, PA*
3.82/4.0 GPA
Master's in Robotics, BSE in Mechanical Engineering | Expected Graduation: May 2029
Relevant Coursework: Static Mechanics, Thermal Fluids, Linear Algebra, Multivariable Calculus, Differential Equations

EXPERIENCE

Penn Electric Racing (FSAE) | Philadelphia, PA — Sept. 2025 – Present
HV Battery Pack Responsible Engineer

- Led mechanical design and manufacturing of **high-voltage tractive battery pack** and **battery management enclosure**, utilizing welded aluminum construction and 3D-printed ULTEM components.
  - Accommodated transition from **RWD to 4WD**, integrating increased battery capacity and mass while maintaining structural integrity and chassis compatibility.
  - Reduced HV battery pack mounting tab mass by **62.1%** (287 g) through FEA-driven geometric optimization, achieving a minimum safety factor of 1.07 under FSAE-prescribed 40g lateral and 20g vertical load cases.
  - Redesigned battery management enclosure to increase internal volume by **88.7%** (7.05 L) to improve component accessibility, cable routing, and serviceability.
- Developed **custom vibration test fixture** capable of 334% higher theoretical excitation frequency than previous design, improving stability and constraining motion through linear bearings.
- Designed and manufactured **battery charge cart**, suspension mounting tabs, and 15 manufacturing jigs to support vehicle assembly and testing, also helped run suspension control arm compression testing.

ModLab | Pennovation Works | Philadelphia, PA — May 2026 - Present
Undergraduate Researcher

- Designed and fabricated **custom fatigue testing fixture** for twisted-string actuators for a self-configurable modular robot, enabling repeatable lifecycle testing.
- Redesigned actuator with **bearing integration** and **symmetric geometry** about the string plane, increasing fatigue life from 10,000 to 50,000 cycles (**5× improvement**).
- Acknowledged for technical contributions in research paper submitted to IEEE ICRA 2027.
- Designing and prototyping **docking mechanism** for Variable Topology Truss (VTT), a reconfigurable robotic structure.

Precision Machining Laboratory | University of Pennsylvania MEAM Labs — August 2026 - Present
Machine Lab Assistant

- Instruct engineering students in precision machining of Stirling engine components using manual mills and lathes, including facing, drilling, tapping, reaming, boring, and parting operations.
- Provide technical guidance on machining setups, CNC programming with Mastercam and ProtoTRAK, and dimensional inspection using precision metrology tools.

PROJECTS

Stirling Engine | University of Pennsylvania | Philadelphia, PA — Spring 2026

- Designed, precision machined, and assembled custom Spiderman-themed Stirling engine using manual mills and lathes, achieving operational speeds of 500 RPM.

Polaris (GPS-Based Anchored Compass) | University of Pennsylvania | Philadelphia, PA — Spring 2026

- Developed GPS-based electronic compass to point back to user-defined waypoints, integrating Arduino, GPS, IMU, and LED matrix for real-time directional feedback and distance information.

Analyzing Drag Coefficients of 3D Shapes at Terminal Velocity | PGSS, CMU | Pittsburgh, PA — Summer 2024

- Built a wind tunnel out of office supplies and researched the effects of wing shape on drag.
- Led data analysis and error discussion, verified corresponding drag coefficients within 97% of NASA's accepted values.

SKILLS

Software: CAD (SolidWorks, AutoDesk, Onshape), Finite Element Analysis (FEA), Arduino IDE, Python, Java, OCaml
Hardware: Milling, Laser-cutting, 3D printing, GD&T, Tolerance Stack-up, Circuits
Languages: Spanish, intermediate proficiency